IBM University Engagement Project Submission

"Exploring the Power of Machine Learning with IBM Cloud"

Project Tile – Rural Infrastructure Project Classification Using Machine Learning

Domain of Project – Rural Development / Infrastructure

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Problem Statement

Problem Statement: Rural Infrastructure Project Classification Using Machine Learning

Rural infrastructure development plays a vital role in improving transportation, education, healthcare, and economic growth. Government projects are implemented under different PMGSY (Pradhan Mantri Gram Sadak Yojana) schemes based on project characteristics. Manually classifying these infrastructure projects is time-consuming and prone to errors.This project aims to develop a Machine Learning model that automatically classifies rural infrastructure projects into the appropriate PMGSY category using historical project data. The solution helps improve decision-making, reduces manual effort, and supports efficient project management.

Dataset link :

https://aikosh.indiaai.gov.in/web/datasets/details/pradhan_mantri_gram_sadak_yojna_pmgsy.html

Proposed Solution - IBM watsonx.ai Studio – AutoAI

This project uses IBM watsonx.ai Studio AutoAI to automate the classification of rural infrastructure projects.

The solution performs:

- Data preprocessing
- Automatic feature engineering
- Machine Learning model selection
- Hyperparameter optimization
- Model evaluation

The trained model predicts the correct PMGSY project category with high accuracy, enabling faster and more reliable classification.

Technology Used

Platform

- IBM Cloud
- IBM watsonx.ai Studio
- IBM AutoAI

Programming Language

- Python

Machine Learning Type

- Classification

Algorithm Used

- Snap Random Forest Classifier
- XGBoost Classifier

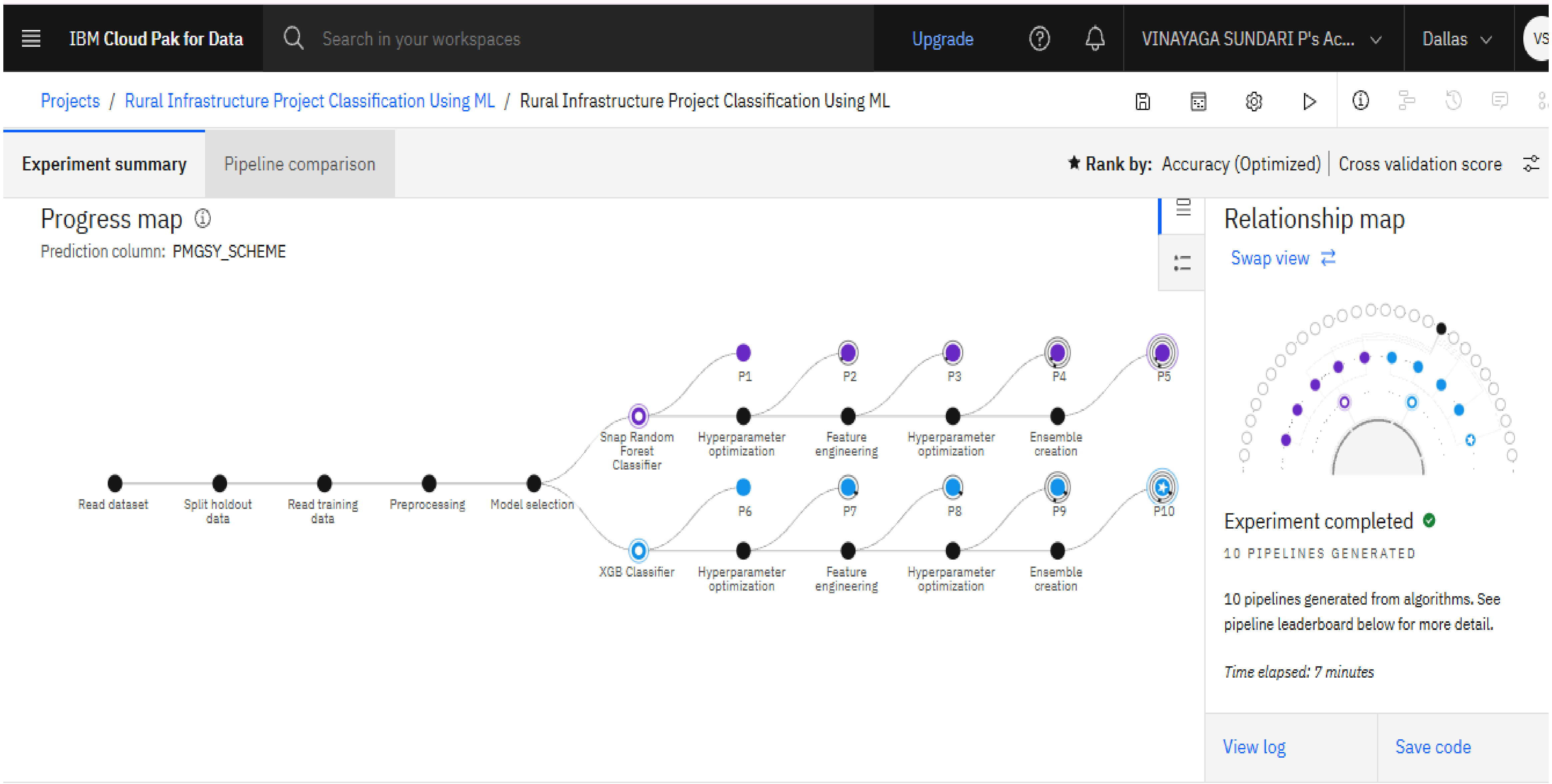
Technology Used

Dataset - PMGSY_DATASET.csv
https://aikosh.indiaai.gov.in/web/datasets/details/pradhan_mantri_gram_sadak_y ojna_pmgsy.html

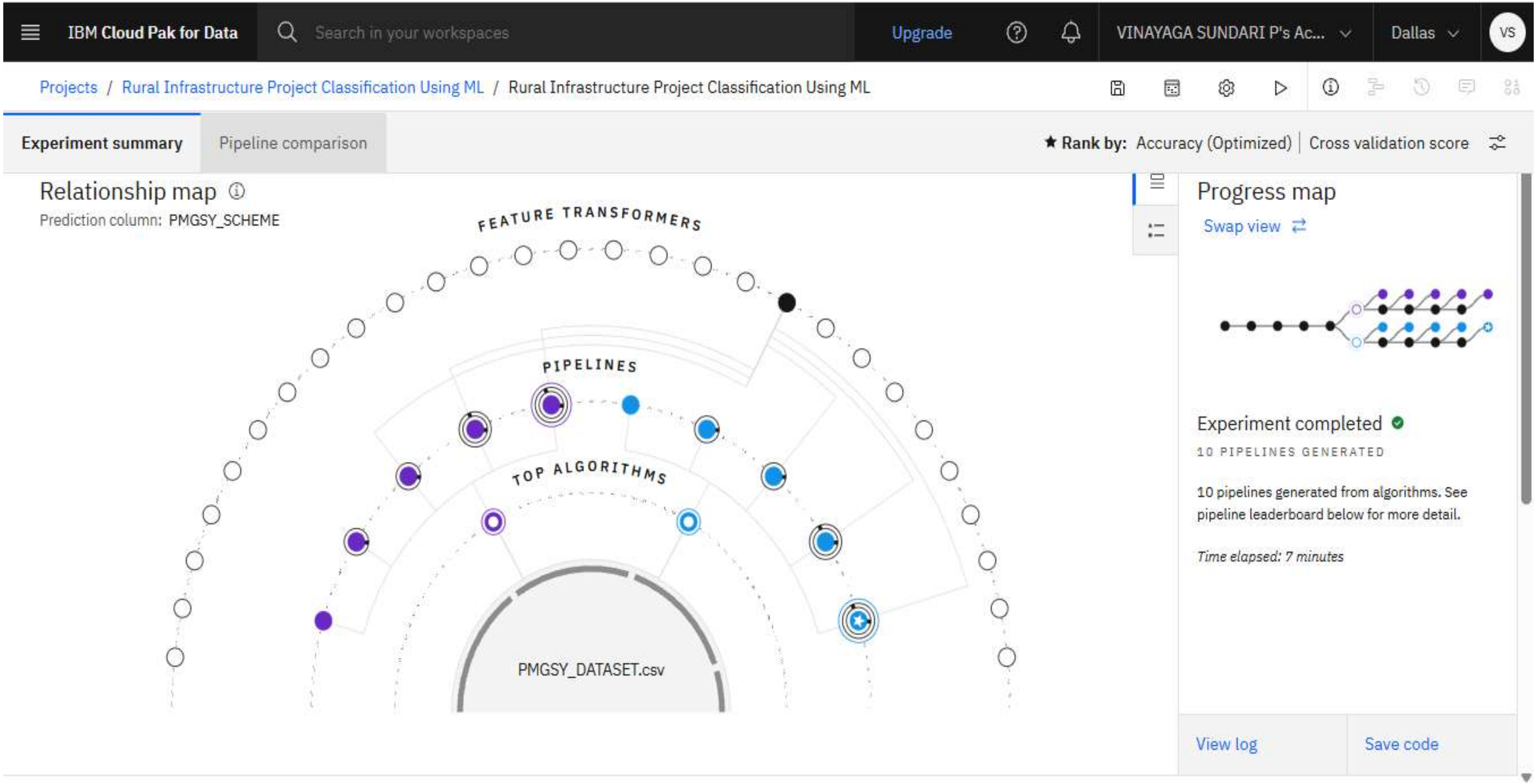
Accuracy Obtained :
The Snap Random Forest Classifier was selected as the best model with an accuracy of **92.4%**.

Hyperparameter Tuning :
Automatically optimized using IBM AutoAI

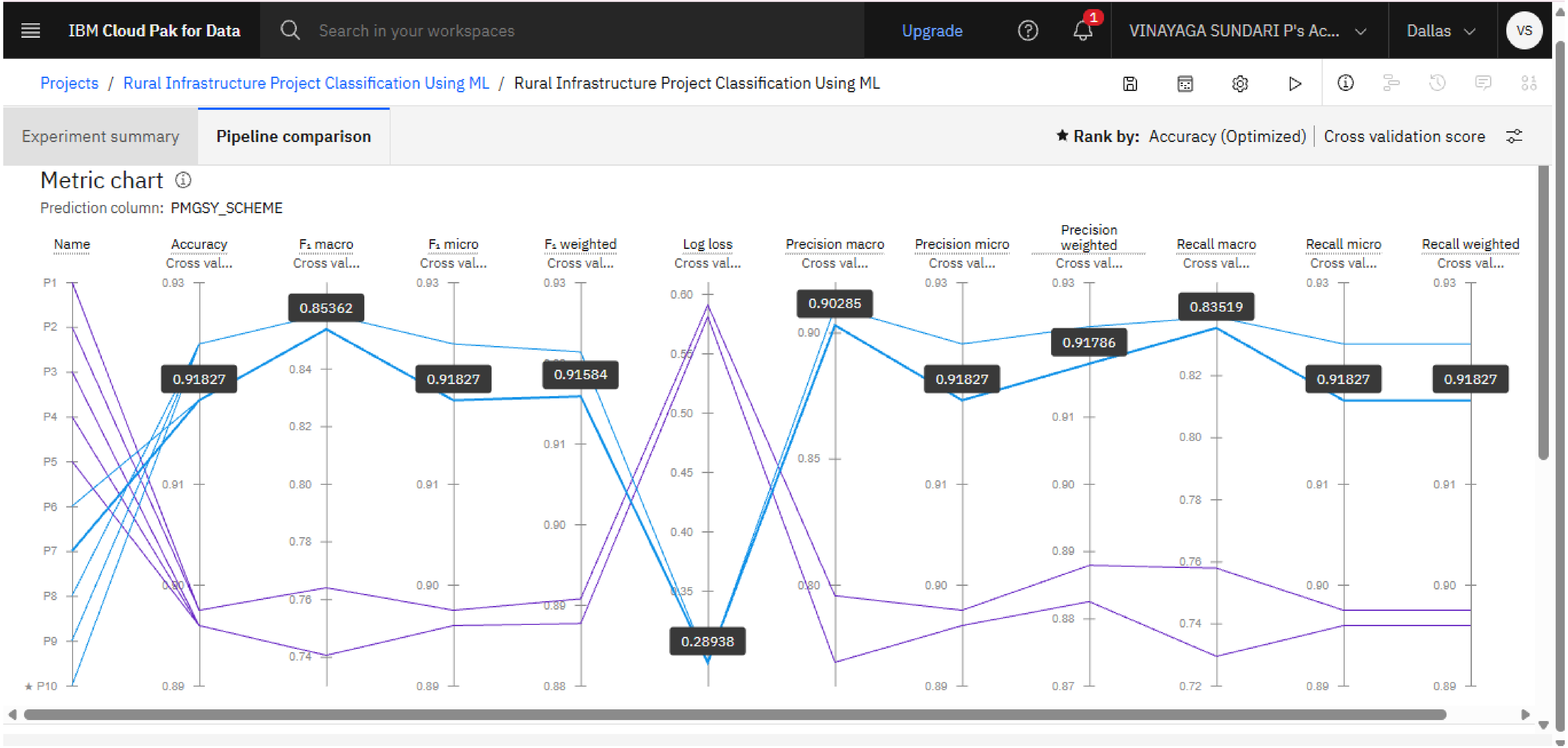
Training Screenshots-



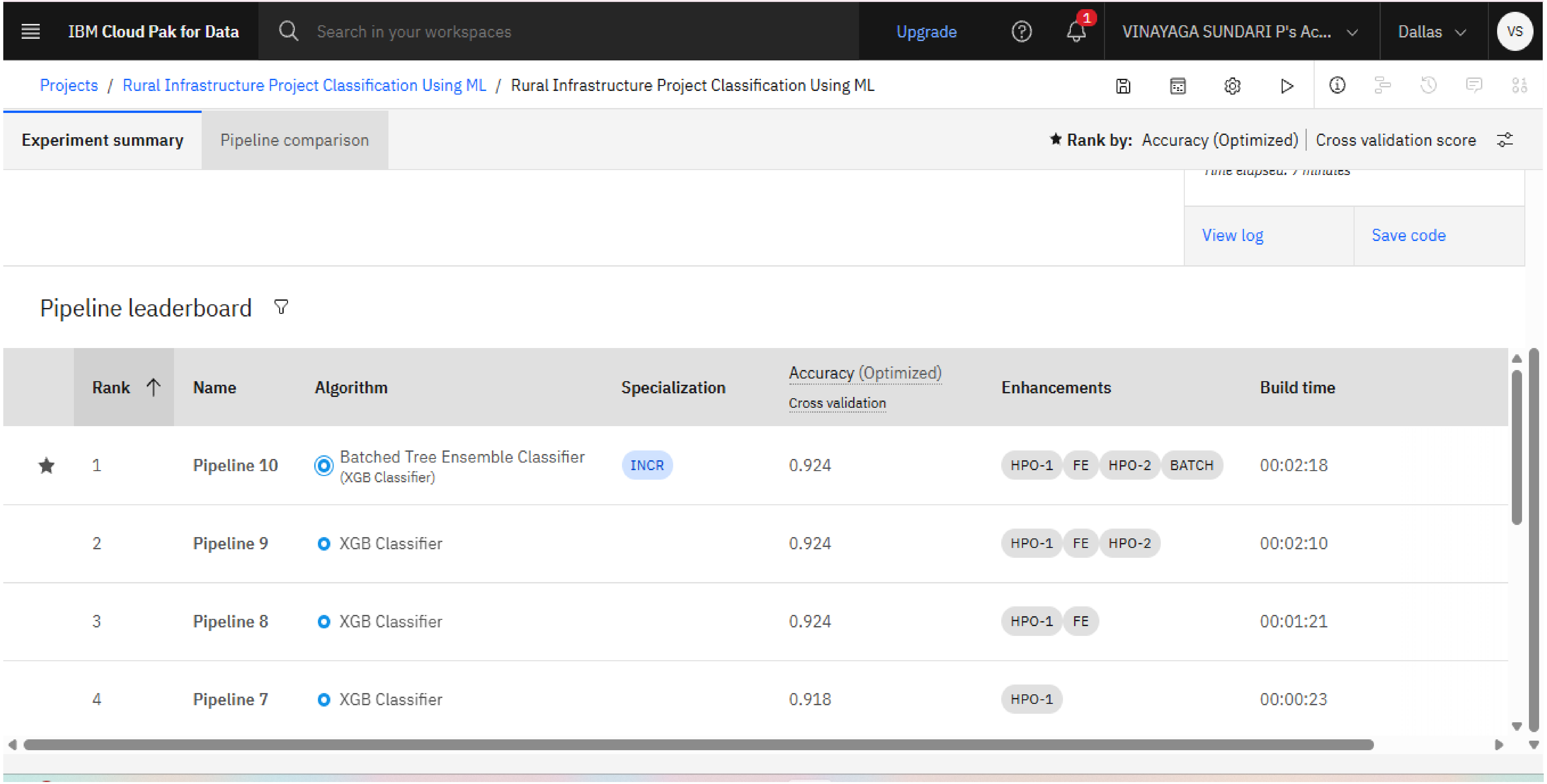
Training Screenshots-



Training Screenshots-

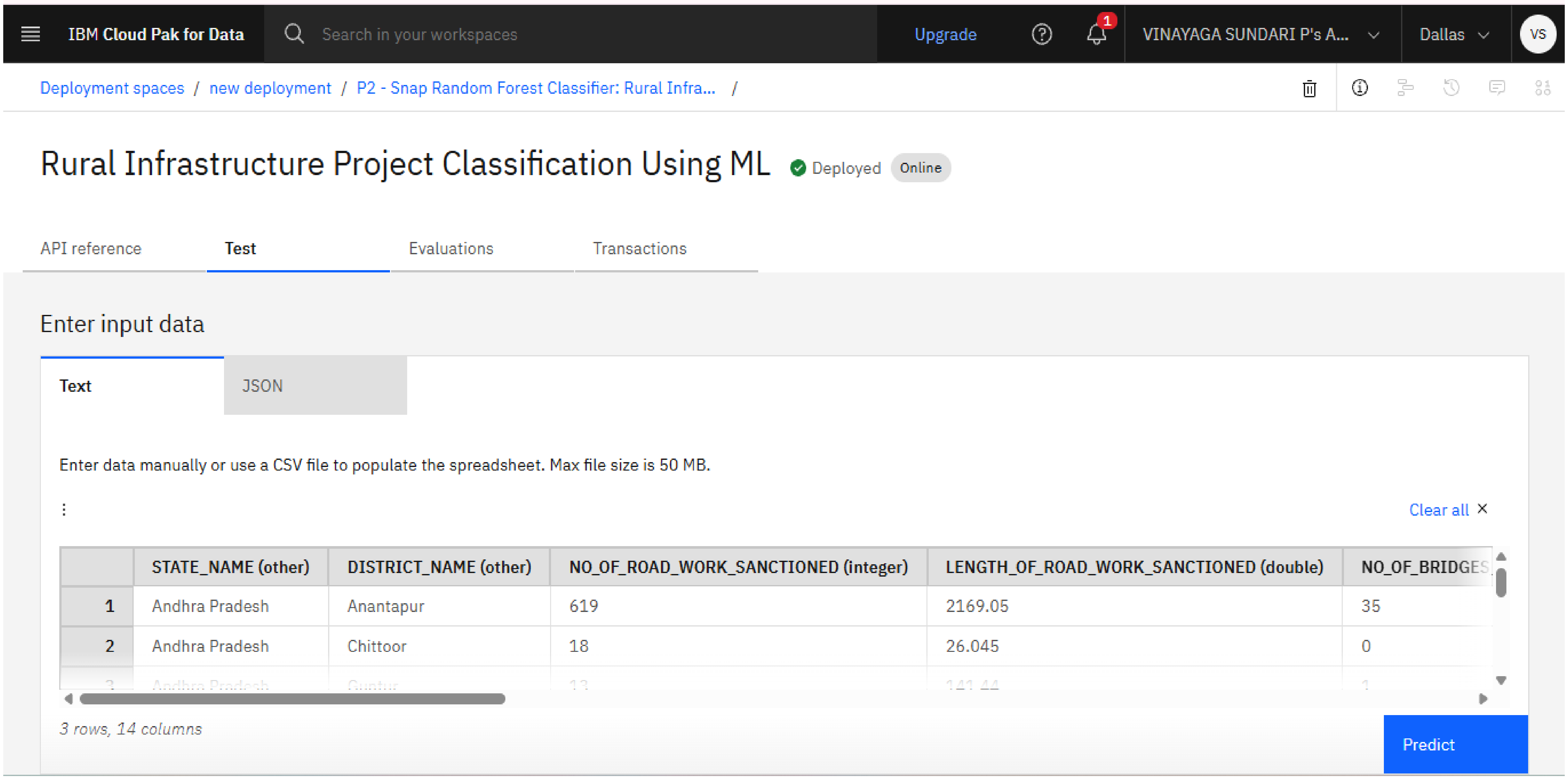


Best Model Screenshot:

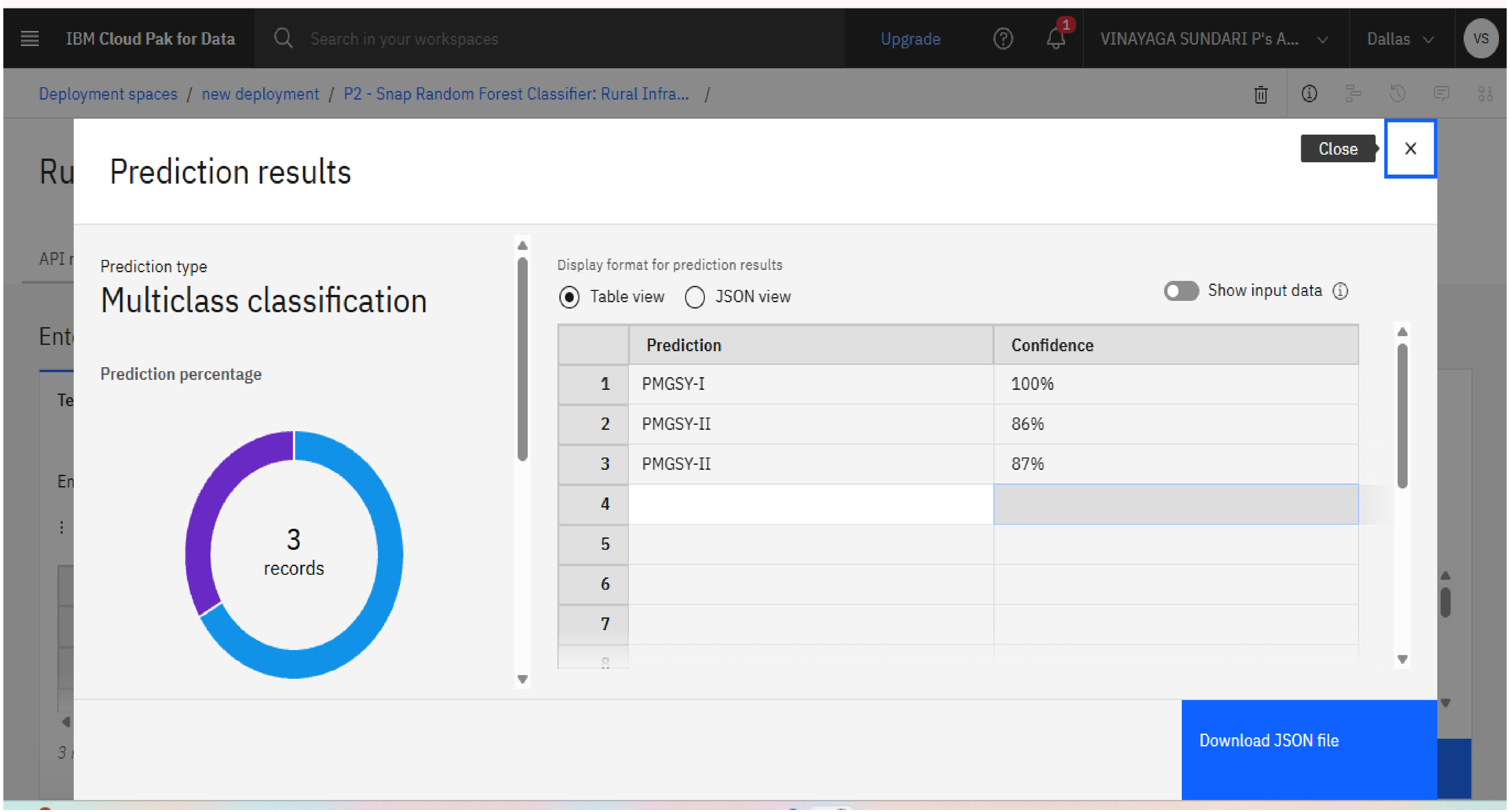


	Rank	Name	Algorithm	Specialization	Accuracy (Optimized) Cross validation	Enhancements	Build time
★	1	Pipeline 10	Batched Tree Ensemble Classifier (XGB Classifier)	INCR	0.924	HPO-1 FE HPO-2 BATCH	00:02:18
	2	Pipeline 9	XGB Classifier		0.924	HPO-1 FE HPO-2	00:02:10
	3	Pipeline 8	XGB Classifier		0.924	HPO-1 FE	00:01:21
	4	Pipeline 7	XGB Classifier		0.918	HPO-1	00:00:23

Testing Screenshot:



Testing Screenshot:



Novelty and Uniqueness

- Automates rural infrastructure project classification.
- Reduces manual classification errors.
- Uses IBM AutoAI for automatic model generation.
- Provides fast and accurate predictions.
- Supports government planning and rural development initiatives.
- Easily scalable for larger infrastructure datasets.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository) with format –

Please test and Paste the working Github URL –

Make sure you have Uploaded following three files into your github repository

1) Data.csv file, 2) yourproblemstatement.pdf file 3) your project presntation.pptx file 4) Python Notebook downloaded from IBM Cloud

Future Scope

- Improve prediction accuracy using larger datasets.
- Integrate GIS and satellite data for infrastructure planning.
- Develop a web application for real-time project classification.
- Deploy the model as a cloud-based API using IBM Cloud.

·Extend the solution to classify additional government infrastructure schemes.

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Completion date: 19 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.